

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and Communications Technology

Use Case Specification

Pay Order by Credit Card (Customer / PayPal)

AIMS – An Internet Media Store

ITSS Software Development

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Use Case “Pay Order by Credit Card”

1. Use case code

UC003

2. Brief Description

This use case describes the interaction between the Customer and the AIMS Software, and between AIMS Software and PayPal Sandbox when the Customer wishes to pay for their order using a credit card via the PayPal payment gateway. This use case is an alternative payment flow within the Place Order use case (UC002), triggered when the customer chooses credit card as the payment method instead of the default QR code.

3. Actors

3.1 Customer

The end user who browses products, adds items to the cart, and initiates payment.

3.2 AIMS Software

The e-commerce system that orchestrates the payment flow.

3.3 PayPal Sandbox

External payment service that processes credit card transactions and refunds (sandbox/testing environment).

4. Preconditions

1. The Customer has a valid cart with at least one item.
2. All items in the cart have sufficient stock.
3. The Customer has entered and confirmed valid delivery information.
4. The AIMS Software has displayed the invoice screen with temporary order information.
5. The Customer chooses to switch from the default QR code payment to credit card payment.

5. Basic Flow of Events

Step 1. AIMS Software displays the credit card payment form (see Table A).

Step 2. Customer enters credit card information and submits the form.

Step 3. AIMS Software validates the input fields [is valid = true].

Step 4. AIMS Software sends payment request to PayPal (amount, currency, card info, order reference).

Step 5. PayPal processes the credit card payment.

Step 6. PayPal returns payment result to AIMS Software.

Step 7. AIMS Software checks whether the payment is successful [is successful = true].

Step 8. AIMS Software records payment transaction information.

Step 9. AIMS Software saves the successfully paid order (status = Pending Processing).

Step 10. AIMS Software empties the cart.

Step 11. AIMS Software performs the following two actions in parallel:

- Displays the order confirmation screen.
- Sends the invoice and transaction information to the Customer's email.

Step 12. AIMS Software displays the transaction information (see Table B).

Step 13. Customer views the order confirmation screen.

Step 14. Customer decides whether to cancel the order [cancel = no] → use case ends normally.

6. Alternative flows

N o	Location	Condition	Action	Resume location
1	Step 3	Customer enters invalid or incomplete credit card information	AIMS Software highlights invalid or missing fields and notifies the Customer.	Step 2
2	Step 6	PayPal returns a payment failure response [is	AIMS Software displays the payment failure reason to the Customer. Customer decides:	Step 2 (retry) or Use case ends (cancel)

No	Location	Condition	Action	Resume location
		successful = false]	• [retry = true]: Return to Step 2 to re-enter credit card information. • [retry = false]: Customer selects Cancel; use case ends.	
3	Steps 1–4 (before payment submission)	Customer cancels or exits the payment process before submitting	1. Customer clicks “Cancel” or closes the PayPal interface. 2. AIMS Software returns the Customer to the invoice / payment-selection screen. 3. No payment is charged; the order is NOT finalised.	Use case ends
4	Step 13 (after order confirmation displayed)	Customer chooses to cancel the order after successful payment [cancel = yes]	1. Customer confirms the cancellation request. 2. AIMS Software sends a full refund request to PayPal; PayPal processes the refund automatically. 3. AIMS Software updates the order status to “Cancelled” and sends a cancellation and refund notification to the Customer’s email. 4. Customer receives the cancellation and refund confirmation.	Use case ends

7. Input data

Table A – Input data of Pay Order by Credit Card

N o	Data Fields	Description	Ma nda tory	Valid Condition	Example
1	Cardholder Name	Full name as printed on the credit card	Yes	Non-empty string; letters and spaces only	Nguyen Tuan Dat
2	Card Number	16-digit credit card number	Yes	16 digits numeric	4111111111111111
3	Expiration Date	Month/Year expiry date on the card	Yes	MM/YY format; must be a future date	12/27
4	CVV / CVC	3 or 4 digit security code on back of card	Yes	3–4 numeric digits	123
5	Billing Address	Address associated with the credit card account	No	Free text; max 200 characters	Dong Da, Ha Noi

8. Output data

Table B – Output data of Pay Order by Credit Card

N o	Data Fields	Description	Display Format	Example
1	Customer Name	Full name of the customer who placed the order	Plain text	Nguyen Tuan Dat
2	Phone Number	Customer's contact phone number	Plain text (10–11 digits)	0385388626
3	Shipping Address	Delivery address entered by the customer	Plain text	Luong Dinh Cua
4	Province / City	Destination province or city	Plain text	Ha Noi

N o	Data Fields	Description	Display Format	Example
5	Total Amount	Total amount paid including VAT and delivery fee	Currency (VND)	1,200,000 VND
6	Transacti on ID	Unique identifier returned by PayPal for this transaction	Alphanumeric string	PAYID-123456
7	Transacti on Content	Description / note attached to the transaction	Plain text	Mua hang
8	Transacti on Datetime	Date and time the transaction was completed	DD/MM/YYYY Y HH:MM:SS	15/03/2026 10:30:18

9. Postconditions

6. The payment transaction is successfully processed and recorded by the AIMS Software.
7. The order is saved with status "Pending Processing".
8. The Customer receives a confirmation email containing invoice details and transaction information.
9. The cart is cleared after successful order placement.